



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

REVIEW TEST

15-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C,D	A,D	A,B,C	A,C	A,B	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	D	A	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	4.00	2.44	0.76 to 0.78	1.22	9.80	0.08 to 0.09	4.00	0.49 to 0.50

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	D	B,C,D	A,C	A,B,D	A,C	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	C	C	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	6.00	2.00	0.80	5.00	4.00	7.00	5.00	8.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,D	A,B,C	A,D	A,B,C	B,C,D	C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	D	D	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.42 to 5.44	3.66 to 3.67	1.50	8.00	0.40	1.01	0.53	0.75

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,C,D)

$$(A) F = \frac{2IA}{C} = \frac{2 \times 200 \times 0.2 \times 0.3}{3 \times 10^8} = 0.08 \mu\text{N}$$

$$(C) F = \frac{IA \cos \theta}{C} = \frac{200 \times 0.2 \times 0.3}{3 \times 10^{-8}} \times \frac{\sqrt{3}}{2} = 2\sqrt{3}$$

$$(D) F = \frac{2IA \cos^2 \theta}{C} = 0.06 \mu\text{N}$$

2. Ans (A,D)

$$y_1 = 5.5 \frac{\lambda D}{d} = (\mu - 1) t \frac{D}{d}$$

$$y_2 = 3.5 \frac{\lambda D}{\mu_0 d} = \left(\frac{\mu}{\mu_0} - 1 \right) t \frac{D}{d}$$

$$\frac{11}{7} = \frac{(\mu - 1)}{(\mu - \frac{4}{3})}$$

$$11\mu - \frac{44}{3} = 7\mu - 7$$

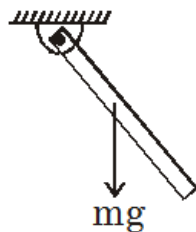
$$4\mu = \frac{44}{3} - 7 = \frac{23}{3}$$

$$\mu = \frac{23}{12}$$

$$5.5 \times 600 \times 10^{-9} = \frac{11}{12} t$$

$$t = 3.6 \mu\text{m}$$

6. Ans (A,C)



$$I\alpha = \tau_{\text{ext}} = -mg\frac{l}{2}\sin\alpha + \tau$$

$$\alpha + \frac{\frac{mg\ell}{2}}{\frac{m\ell^2}{3}}\theta = \frac{0.5}{\frac{m\ell^2}{3}}\sin 2t$$

$$\alpha + \frac{30}{2 \times 3}\theta = \frac{0.5}{1 \times 3} \times 3 \sin 2t$$

$$\alpha + 5\theta = \frac{1}{6} \times \sin 2t$$

$$\theta = \theta_0 \sin 2t$$

$$\alpha = 4\theta_0 \sin 2t$$

$$5\theta_0 \sin 2t - 4\theta_0 \sin 2t = \frac{1}{6} \times \sin 2t$$

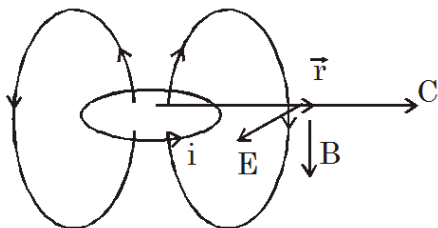
$$\theta_0 = \frac{1}{6} \text{ rad}$$

same phase.

PART-1 : PHYSICS

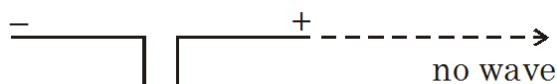
SECTION-I (ii)

7. Ans (B)



Electric dipole antenna should be parallel to $\vec{E}$

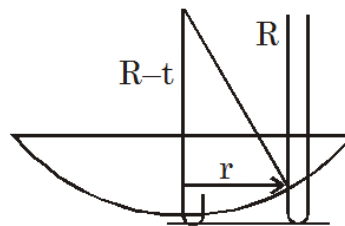
8. Ans (D)



9. Ans (A)

Centre is bright & monochromatic fringes not coloured fringes.

10. Ans (A)



Centre is maxima

$$2 \times 1.4 \times t = 5\lambda$$

$$2Rt = r^2$$

$$r = \sqrt{2Rt}$$

$$\sqrt{2R \times \frac{5\lambda}{2.8}}$$

$$= \sqrt{\frac{2 \times 1 \times 5 \times 6.3 \times 10^{-8}}{2.8}}$$

$$= 1.5 \times 10^{-3} \text{ m}$$

PART-1 : PHYSICS

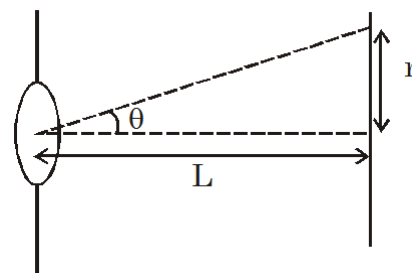
SECTION-II

1. Ans (4.00)

$$E = \frac{n^2 h^2}{8mL^2} = \frac{1}{2}mv^2$$

$$\Rightarrow n = \frac{2mvL}{h} = \frac{2 \times 6.63 \times 10^{-15} \times 10^{-19} \times 2}{6.63 \times 10^{-34}} = 4$$

2. Ans (2.44)



$$\sin \theta = \frac{1.22\lambda}{D}$$

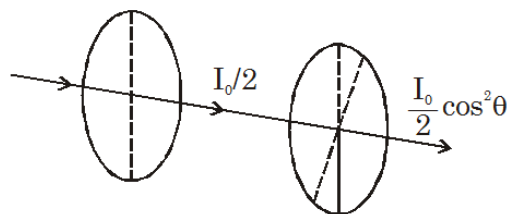
$$\frac{r}{L} = \frac{1.22h}{mvD}$$

$$r = \frac{1.22hL}{mvD}$$

$$= \frac{1.22 \times 6.4 \times 10^{-34} \times 1}{1.6 \times 10^{-27} \times 10^3 \times 2 \times 10^{-7}}$$

$$= 2.44 \text{ mm}$$

3. Ans (0.76 to 0.78)



$$\frac{I_0}{2} \cos^2 \theta = \frac{3I_0}{10}$$

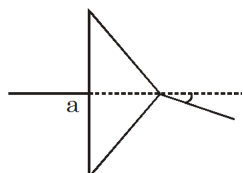
$$\cos \theta = \sqrt{\frac{3}{5}} = \frac{\sqrt{3}}{\sqrt{5}} = \frac{1.73}{2.24} = 0.77$$

4. Ans (1.22)

$$\theta_{\min} = \frac{1.22\lambda}{D} = \frac{1.22 \times 5 \times 10^{-7}}{5}$$

$$= 1.22 \times 10^{-7} \text{ rad.}$$

5. Ans (9.80)



$$\beta = \frac{0.5}{20} = \frac{1}{40} \text{ cm}$$

$$d = 2a\delta = 2a(\mu - 1)\alpha$$

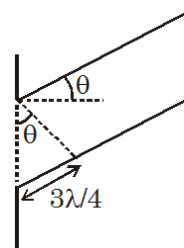
$$D = a + b = 5a$$

$$\frac{1}{4000} = \beta = \frac{\lambda D}{d} = 490 \times 10^{-9} \times \frac{5a}{2a(0.5)\alpha}$$

$$\alpha = 98 \times 10^{-4} \text{ rad.}$$

$$= 9.8 \text{ m rad.}$$

6. Ans (0.08 to 0.09)



$$\beta = \frac{\pi \omega \sin \theta}{\lambda} = \frac{3\pi}{4}$$

$$\omega \sin \theta = \frac{3\lambda}{4}$$

$$I = I_0 \frac{\sin^2 \beta}{\beta^2}$$

$$\frac{I}{I_0} = \frac{\sin^2 \frac{3\pi}{4}}{\frac{9\pi^2}{16}} = \frac{16}{9\pi^2 \times 2} \approx \frac{8}{90}$$

7. Ans (4.00)

$$P_{\text{final}} = \frac{P_1 V_1 + P_2 V_2}{V_1 + V_2} = \frac{2 + 6}{5} = \frac{8}{5} \text{ atm}$$

8. Ans (0.49 to 0.50)

$$A = A_0 e^{\frac{-bt}{2m}}$$

$$0.37 = e^{\frac{-bt}{2m}}$$

$$\frac{bt}{2m} = 1 \Rightarrow b = \frac{2 \times 1}{4} = 0.5 \text{ kg/sec}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (D)

(A) Ionisation isomer of $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})\text{Cl}_2]\text{Br}$ is $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})\text{ClBr}]\text{Cl}$ and its geometrical isomers are 4

(B) $d_{\text{C-O}}$ (bridging) $>$ $d_{\text{C-O}}$ (terminal)

(C) Di- μ -chloridotetrakis(trichlorostannito)dirhodate(I) ion

(D) $[\text{Co}(\text{NO}_2)_6]^{4-} = d^2 sp^3$
 $[\text{CoF}_6]^{3-} = sp^3 d^2$

2. Ans (B,C,D)

Option (B) is true for Li_2 and F_2

Option (C) is true for C_2

Option (D) is true for B_2

4. Ans (A,B,D)

$$(N \times V)_{\text{I}_2} = (2 \times 1) \times 250 = 500$$

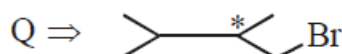
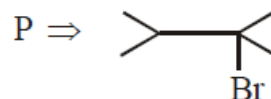
$$\text{n. f. for } \text{Cu}_2\text{S} = 10, N \times V = (0.5 \times 10) \times 100 = 500$$

$$\text{n. f. for } \text{CuS} = 8, N \times V = (0.5 \times 8) \times 125 = 500$$

$$\text{n. f. for } \text{S}_2\text{O}_3^{2-} = 1, N \times V = (0.5 \times 1) \times 1000 = 500$$

$$\text{n. f. for } \text{I}_2 = 2$$

5. Ans (A,C)

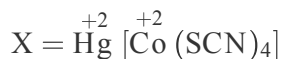


PART-2 : CHEMISTRY

SECTION-I (ii)

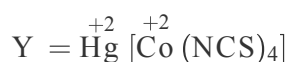
7. **Ans (C)**

$\text{HgI}_2 \rightarrow$ scarlet red precipitate



hyb = sp^3

n = 3



hyb = sp^3

n = 3

8. **Ans (C)**

$\text{Hg}[\text{Co}(\text{SCN})_4]$		$\text{Hg}[\text{Co}(\text{NCS})_4]$	
$(\text{SCN})^-$	$(\text{NCS})^-$	$(\text{NCS})^-$	$(\text{SCN})^-$
4	0	4	0
3	1	3	1
2	2	2	2
1	3	1	3
0	4	0	4
total = 5		total = 5	

sum = 5+5 = 10

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (6.00)**

Option (R) and (W) are incorrectly matched.

2. **Ans (2.00)**

(1) It is an endothermic reaction

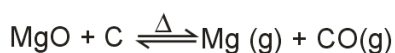
(2) Ilmenite = $\text{FeO} \cdot \text{TiO}_2$

(3) KCN act as depressant

(4) $\text{Zn} \rightarrow$ distillation

(5) Complex formed is $\text{Na}[\text{Au}(\text{CN})_2]$

(6) Leaching of red bauxite



(7) $\downarrow$ Rapid cooling
 $\text{Mg} (\text{s}) + \text{CO} (\text{g})$

3. **Ans (0.80)**

$$A_1 = \pi \sigma_{xy}^2 \cdot \sqrt{\frac{8KT}{\mu_{xy}}} \cdot N_A^2 \cdot P$$

$$\left(\frac{A_2}{A_1} \right)^2 = \frac{\mu_{XY}}{\mu_{MN}} = \frac{4}{5} = 0.80$$

4. **Ans (5.00)**

(i), (ii), (iii), (iv), (ix)

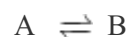
5. **Ans (4.00)**

$$E = E^\circ - 0.06 \text{ pH}$$

$$0.22 = 0.46 - 0.06 \text{ pH}$$

$$0.06 \text{ pH} = 0.46 - 0.22$$

6. **Ans (7.00)**



$$t = 0 \quad 21 \quad -$$

$$t = t_{\text{eq.}} \quad 21-x \quad x$$

$$\frac{x}{21-x} = \frac{4 \times 10^{-2}}{2 \times 10^{-2}} = 2$$

$$x = 14$$

equilibrium concentration of

$$A = 21 - x = 21 - 14$$

$$= 7 \text{ M}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,D)**

$$\lim_{\alpha \rightarrow 0} \frac{\alpha^2}{(\sec \alpha)^{1/x} - (\sec \alpha)^{1/y}} = \frac{0}{0} \text{ form}$$

apply L-Hopital Rule

$$\lim_{\alpha \rightarrow 0} \frac{2\alpha}{\frac{1}{x} (\sec \alpha)^{1/x} \tan \alpha - \frac{1}{y} (\sec \alpha)^{1/y} \tan \alpha}$$

$$f(x, y) = \frac{2}{\frac{1}{x} - \frac{1}{y}} = \frac{2xy}{y-x}$$

$$\Rightarrow f(2, 3) = 12, \frac{2xy}{y-x} = 12 \Rightarrow xy = 6y - 6x$$

$$y = \frac{6x}{6-x}$$

$$\text{differentiate both sides } \frac{dy}{dx} = 6 \left\{ \frac{(6-x)+x}{(6-x)^2} \right\}$$

$$\frac{dy}{dx} = \frac{36}{(6-x)^2} \Rightarrow \frac{dy}{dx} \Big|_{x=2} = \frac{36}{16} = \frac{9}{4}$$

$\frac{36}{(6-x)^2}$ can't be zero $\Rightarrow$ No horizontal tangent

2. Ans (A,B,C)

$$\int \cot \left(2 \tan^{-1} \sqrt{\frac{\sqrt{1+\sqrt{x}} - x^{1/4}}{\sqrt{1+\sqrt{x}} + x^{1/4}}} \right) dx$$

$$x = \tan^4 \theta$$

$$\int \cot \left(2 \tan^{-1} \sqrt{\frac{\sec \theta - \tan \theta}{\sec \theta + \tan \theta}} \right) dx$$

$$\int \cot \left(2 \tan^{-1} \sqrt{\frac{\left(\sin \frac{\theta}{2} - \cos \frac{\theta}{2} \right)^2}{\left(\cos \frac{\theta}{2} + \sin \frac{\theta}{2} \right)^2}} \right) dx$$

$$\int \cot \left(2 \tan^{-1} \tan \left(\frac{\pi}{4} - \frac{\theta}{2} \right) \right) dx$$

$$= \int \cot \left(\frac{\pi}{2} - \theta \right) = \int \tan \theta d\theta = \int x^{1/4} dx$$

$$= \frac{4x^{5/4}}{5} + C$$

3. Ans (A,D)

$$y^x - x^y = 1$$

$$\text{at } x = 1 \Rightarrow y = 2$$

$$\text{differentiate}$$

$$x \cdot y^{x-1} \cdot \frac{dy}{dx} + y^x \cdot \ln y - y \cdot x^{y-1} - x^y \cdot \ln x \cdot \frac{dy}{dx} = 0$$

$$\text{put } x = 1 \text{ \& } y = 2$$

$$\frac{dy}{dx} + 2 \ln 2 - 2 = 0$$

$$\Rightarrow \frac{dy}{dx} = 2(1 - \ln 2)$$

$$\Rightarrow \ln \left(\frac{e^2}{4} \right)$$

5. Ans (B,C,D)

$$I = \int \frac{x + \sqrt[3]{x^2} + \sqrt[6]{x}}{x(1 + \sqrt[3]{x})} dx \text{ Put } x = t^6$$

$$dx = 6t^5 dt$$

$$I = \int \frac{t^6 + t^4 + t}{t^6(1 + t^2)} \cdot 6t^5 dt = 6 \int \frac{1 + t^3 + t^5}{(1 + t^2)} dt$$

$$= 6 \left(\int \frac{dt}{1+t^2} + \int t^3 dt \right) = \frac{6t^4}{4} + 6t \tan^{-1} t + c = \frac{3x^{2/3}}{2} + 6 \tan^{-1} (x^{1/6}) + c$$

$$\therefore A = \frac{3}{2}, B = \frac{2}{3}, C = 6, D = \frac{1}{6} \Rightarrow AB = 1,$$

$$AB + C = 7$$

$$\text{Now, } \left(AB + \frac{x}{D} \right)^c = (1 + 6x)^6$$

$$\therefore \text{sum of all coefficients of even power of } x$$

$$= {}^6C_0 + {}^6C_2 \cdot 6^2 + {}^6C_4 \cdot 6^4 + {}^6C_6 \cdot 6^6$$

$$= \frac{(1+6)^6 + (1-6)^6}{2} = \frac{(1+c)^c + (c-1)^c}{2}$$

6. Ans (C)

$$\frac{f'(x)}{f(x)} = \frac{{}^nC_0}{{}^nC_0 x + 1} + \frac{{}^nC_1}{{}^nC_1 x + 2} + \dots + \frac{{}^nC_n}{{}^nC_n x + n + 1}$$

$$\Rightarrow \frac{f'(0)}{f(0)} = \frac{{}^nC_0}{1} + \frac{{}^nC_1}{2} + \dots + \frac{{}^nC_n}{n+1}$$

$$f'(0) = (n+1)! \left(\frac{2^{n+1} - 1}{n+1} \right) = n!(2^{n+1} - 1)$$

$$\therefore n!(2^{n+1} - 1) = \frac{3}{2} 7!$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

$$\text{Let } f(x) = (x-1)^2(ax^2 + bx + c)$$

$$\Rightarrow a + b + c = 2 \text{ \& } f'(c) = -8, f'(2) = 8$$

$$\Rightarrow c = 3, a = 1, b = -2$$

$$\Rightarrow f(x) = (x-1)^2(x^2 - 2x + 3)$$

$$g(x) = 3(x-1)^2 \Rightarrow g'(0) = -6, f'(0) = -8$$

$$\text{Angle of intersection}$$

$$= \tan^{-1} \left(\frac{2}{1+48} \right) = \tan^{-1} \frac{2}{49}$$

8. Ans (D)

$$\text{Length of subtangent} = \left| \frac{y}{dy/dx} \right| = \left| \frac{y}{6(x-1)} \right|$$

$$\left| \frac{3(x-1)^2}{6(x-1)} \right| = 2018 \Rightarrow \frac{x-1}{2} = \pm 2018$$

$$x = 4037 \text{ or } x = -4035$$

9. Ans (D)

$$\therefore \lim_{x \rightarrow \infty} \frac{f(x)}{|x|^3} = 0 \Rightarrow f(x) \text{ is either linear or quadratic}$$

$$\text{Let } f(x) = ax^2 + bx (\because f(0) = 0)$$

$$\therefore \lim_{x \rightarrow \infty} (\sqrt{f(x)} - x) = \lim_{x \rightarrow \infty} \frac{(a-1)x^2 + bx}{x(\sqrt{a + \frac{b}{x}} + 1)} = -1$$

$$\Rightarrow a - 1 = 0 \text{ and } \frac{b}{\sqrt{a} + 1} = -1 \Rightarrow b = -2$$

$$\text{Hence, } f(x) = x^2 - 2x$$

$$\text{Now proceed.}$$

10. Ans (B)

$$\because \lim_{x \rightarrow \infty} \frac{f(x)}{|x|^3} = 0 \Rightarrow f(x) \text{ is either linear or quadratic}$$

$$\text{Let } f(x) = ax^2 + bx \quad (\because f(0) = 0)$$

$$\therefore \lim_{x \rightarrow \infty} (\sqrt{f(x)} - x) = \lim_{x \rightarrow \infty} \frac{(a-1)x^2 + bx}{x(\sqrt{a + \frac{b}{x} + 1})} = -1$$

$$\Rightarrow a - 1 = 0 \text{ and } \frac{b}{\sqrt{a} + 1} = -1 \Rightarrow b = -2$$

$$\text{Hence, } f(x) = x^2 - 2x$$

Now proceed.

PART-3 : MATHEMATICS

SECTION-II

1. Ans (5.42 to 5.44)

$$J = \int \left(\frac{1}{y} + \frac{1}{x} \right) dy = \ln y + \frac{y}{x} + C_1$$

$$I = \int \left(\frac{1}{x} + \frac{y}{x^2} \right) dx = \ln x - \frac{y}{x} + C_2$$

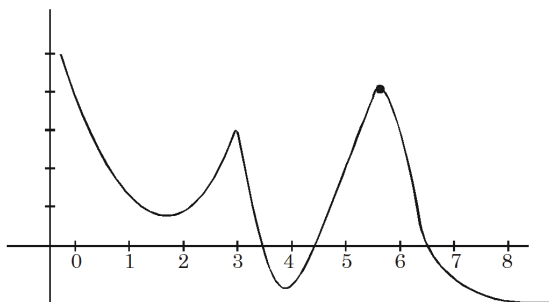
$$g\left(\frac{y}{x}\right) = \ln\left(\frac{y}{x}\right) + \frac{2y}{x} + C$$

$$\frac{y}{x} = 1 \Rightarrow g(1) = \ln(1) + 2 + C \Rightarrow C = -1$$

$$\frac{y}{x} = e \Rightarrow g(e) = \ln e + 2e - 1$$

$$g(e) = 2e = 5.42, 5.43, 5.44$$

2. Ans (3.66 to 3.67)



$$f'(x) = f'(x) (f(x))^2$$

$$\Rightarrow f'(x) = 0, f(x) = 1, f(x) = -1$$

$$f'(x) = 0 \text{ for atleast 4 points}$$

$$f(x) = 1 \text{ for atleast 5 points}$$

$$f(x) = -1 \text{ for atleast 2 point}$$

$$\text{so } f'(x) = f'(x) (f(x))^2 \text{ for atleast 11 points.}$$

3. Ans (1.50)

$$\text{Put } y = 1 : 2 f(x) = f(x) + 3^x \Rightarrow f(x) = 3^x$$

$$\lim_{x \rightarrow 0} \frac{3^x + 3^{-x} - 2}{x^2} = (\ln 3)^2$$

4. Ans (8.00)

$$\lim_{x \rightarrow \infty} \frac{x^3 (\sqrt{1+x^4} - x^2)}{(\sqrt{x^2 + \sqrt{1+x^4} + \sqrt{2}x})}$$

$$\lim_{x \rightarrow \infty} \frac{x^3}{(\sqrt{x^2 + \sqrt{1+x^4} + \sqrt{2}x}) (\sqrt{1+x^4} + x^2)}$$

$$\lim_{x \rightarrow \infty} \frac{1}{\left(\sqrt{1 + \sqrt{\frac{1}{x^4} + 1} + \sqrt{2}} \right) \left(\sqrt{\frac{1}{x^4} + 1} + 1 \right)}$$

$$= \frac{1}{4\sqrt{2}} = \frac{\sqrt{2}}{8}$$

5. Ans (0.40)

$$\text{Let } x = \frac{1}{t}$$

$$\lim_{t \rightarrow 0} \left(\frac{f(3+3t)}{f(3)} \right)^{\frac{1}{t}} = e^{\lim_{t \rightarrow 0} \left(\frac{f(3+3t) - f(3)}{t f(3)} \right)}$$

$$= e^{\lim_{t \rightarrow 0} \left(\frac{f'(3+3t) \cdot 3}{f(3)} \right)} = e$$

6. Ans (1.01)

$$f(x) = \prod_{r=1}^{100} (x-r)^{r(101-r)}$$

$$\ln(f(x)) = \sum_{r=1}^{100} r(101-r) \ln(x-r)$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \sum_{r=1}^{100} \frac{r(101-r)}{(x-r)}$$

$$\therefore \frac{f'(101)}{f(101)} = \sum_{r=1}^{100} \frac{r(101-r)}{(101-r)} = \sum_{r=1}^{100} r$$

$$= \frac{100 \times 101}{2}$$

$$\therefore P = 50 \times 101$$

$$\therefore \frac{P}{5000} = \frac{50 \times 101}{5000} = 1.01$$

7. **Ans (0.53)**

Let width and height of left compartment is x and y then

Volume $V = 3xy$

or $y = \frac{V}{3x}$

$$\therefore \frac{dy}{dx} = \frac{1}{3} \left[\frac{x \frac{dv}{dt} - v \cdot \frac{dx}{dt}}{x^2} \right]$$

$$= \frac{1}{3} \left[\frac{10.5 - 70.3}{10^2} \right] = \frac{-8}{15} \text{ ft/min}$$

$$\left(\frac{dx}{dt} = 3 \text{ ft/min}, \frac{dv}{dt} = 5 \text{ ft}^3/\text{min} \right)$$

 8. **Ans (0.75)**

Put $x = 2, \frac{1}{2}, -1$ successively

$$f(2) + f\left(\frac{1}{2}\right) = 3$$

$$f\left(\frac{1}{2}\right) + f(-1) = \frac{3}{2}$$

$$f(-1) + f(2) = 0$$

solving above, $f(2) = \frac{3}{4}$

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,C,D	A,D	A,B	D	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	D	C	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	270.00 to 275.00	4.00	4.00	4.12	5.00	36.00	2.00	1180.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	B,C	A,C	B,D	A,B	B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	A	A	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	6.00	3.00	1.00	108.00	0.00	2.00	5.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,D	A,B,C,D	A,B,C	B,C	A,B,C	C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	C	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.50	3.20	2.00	0.50	9.00	0.20	8.00	3.75

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A,C)

$$\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0}$$

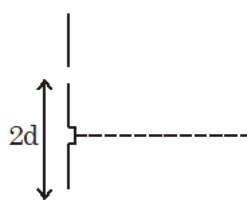
$$B = \frac{E}{C}$$

$$S = \frac{E^2}{C\mu_0} \Rightarrow E = \sqrt{SC\mu_0}$$

$$= \sqrt{300\pi \times 3 \times 10^8 \times 4\pi \times 10^{-7}}$$

$$= 600 \text{ N/C}$$

2. Ans (A,C,D)



$$d \sin \theta = \lambda$$

$$\theta = \frac{\lambda}{d} = 0.6 \times \frac{\pi}{180}$$

$$2d \sin \theta_1 = \lambda$$

$$\sin \theta_1 = \frac{\lambda}{2d}$$

$$d \sin \theta = 4\lambda$$

$$2d \sin \theta = 8\lambda$$

$$2d \sin \theta_2 = \frac{\lambda}{2}$$

$$\sin \theta_2 = \frac{\lambda}{4d} = 0.15^\circ$$

4. Ans (A,B)

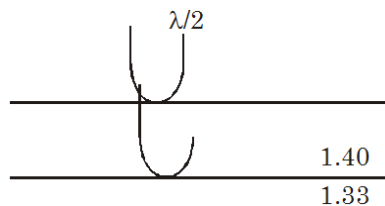
$$k_{\max} = \frac{hC}{\lambda} - \phi = \frac{1240}{500} - 2.2 = 0.28 \text{ eV}$$

$$i_{\text{sat}} = \frac{\eta e I A \lambda}{hC}$$

$$= \frac{A \times 10^{-2} \times 1 \times \frac{1}{2} \times 3 \times 10^8 \times 8.85 \times \frac{10^{-12} \times 10^6}{8.85}}{2.48}$$

$$= \frac{3}{2} \text{ A}$$

5. Ans (D)



$$\Delta x = 2 \times 1.4 \times x = n\lambda$$

$$x = \frac{n \times \lambda}{2.8} = \frac{n \times 560}{2.8} = 200n$$

$$x = m \times \frac{6300}{2.8} = 225m$$

$$x = p \times \frac{7000}{2.8} = 250p$$

6. Ans (A,C)

$$\lambda = \frac{h}{mv}$$

$$v = \frac{eE\tau}{m}$$

$$\lambda = \frac{h}{eE\tau} = \frac{h\ell}{eV\tau}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (D)

$$R = \frac{\epsilon_0 n^2 h^2}{\pi m z e^2}$$

$$v = \frac{ze^2}{2\epsilon_0 nh}$$

$$\frac{1}{\lambda} = \frac{3}{4} \frac{m z^2 e^4}{\epsilon_0^2 h^3 C}$$

$$L = \frac{h}{2\pi}$$

9. Ans (C)

$$(P) \sqrt{\frac{k}{m}} = \omega = 10 \text{ rad/s}$$

$$\gamma = \frac{C}{2m} = \frac{50}{2} = 25$$

over damped.

(Q) As block moves down, spring will be slack.
So undamped SHM.

(R) $mg = kx_0 \Rightarrow$ no motion.

(S) $mg > kx_0$, so it moves down

$$\omega = 5 \text{ rad/s}, \gamma = \frac{C}{2m} = 5 \text{ rad/s}$$

$\Rightarrow$ critically damped.

10. Ans (B)

$$(P) M = \frac{e}{2m} \times \frac{nh}{2\pi}$$

$$(Q) B = \frac{\mu_0 e \omega}{4\pi r} = \frac{\mu_0 e v}{4\pi r^2}$$

$$v = \frac{ze^2}{2\epsilon_0 nh}, r = \frac{\epsilon_0 n^2 h^2}{\pi m z e^2}$$

$$B \propto \frac{z^3}{n^5}$$

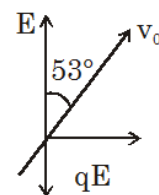
$$(R) \omega = \frac{v}{r} \propto \frac{z^2}{n^3}$$

$$(S) V = -\frac{Ze^2}{4\pi\epsilon_0 r} \propto \frac{z^2}{n^2}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (270.00 to 275.00)



$$v_0 = \frac{h}{m\lambda} = \frac{6.6 \times 10^{-34}}{9 \times 10^{-31} \times 2.2 \times 10^{-7}} = \frac{1}{3} \times 10^4 \text{ m/s}$$

$$53^\circ = \frac{1}{3} \times 10^4 \times \frac{4}{5} = \frac{8}{3} \times 10^3$$

$v_0 \sin \theta \rightarrow$ same

$$v_y = v_0 \cos 53^\circ - \frac{qEt}{m}$$

$$= \frac{1}{5} \times 10^4 - \frac{1.6 \times 10^{-19} \times 9 \times 10^{-9}}{9 \times 10^{-31}}$$

$$= (2 - 1.6) \times 10^3 = 400 \text{ m/s}$$

$$v = \sqrt{400^2 + \left(\frac{8}{3} \times 10^3\right)^2}$$

$$= 400 \sqrt{1 + \frac{400}{9}}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{9 \times 10^{-37} \times 8000} \times 3$$

$$\approx \frac{8000}{3}$$

$$= 275 \text{ nm}$$

2. Ans (4.00)

$$\frac{dH}{dt} = \frac{T - T_0}{\frac{(5a-a)}{4\pi k \times 5a \times a}} = 10 \times C_V \frac{dT}{dt}$$

$$\Rightarrow (T - T_0) 5\pi K a = -10C_V \frac{dT}{dt}$$

$$\int_{3T_0}^{2T_0} \frac{dT}{T - T_0} = \frac{-5\pi K a}{10C_V} \int_0^{t_1} dt$$

$$\ln 2 = \frac{\pi K a}{2C_V} t_1$$

$$t_1 = \frac{2C_V \ln 2}{\pi K a} = \frac{2\pi K a}{\pi K a} = 4 \text{ sec}$$

3. Ans (4.00)

$$\lambda T = C \Rightarrow \lambda \propto \frac{1}{T}; \text{ if energy radiation is}$$

$$\text{constant then } R \propto \frac{1}{T^2} \Rightarrow R \propto \lambda^2$$

$$\frac{R_A}{R_B} = \frac{(\lambda_A)^2}{(\lambda_B)^2}; \frac{R_A}{R_B} = \frac{1}{4}$$

4. Ans (4.12)

$$eV_s = hf - \phi$$

$$v_s = \frac{h}{e} f - \phi$$

$$\text{Slope} = \frac{h}{e} = 4.12 \times 10^{-15}$$

$$h = 4.12 \times 10^{-15} \text{ eVs}$$

7. Ans (2.00)

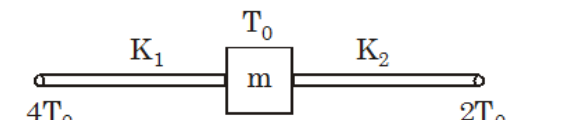
$$KE_{aV} = \frac{3}{2} K_B T$$

$$= \frac{3}{2} \times 1.38 \times 10^{-23} \times 273$$

$$= 5.65 \times 10^{-21} \text{ J}$$

$$6 \times 10^{-21} \text{ J}$$

8. Ans (1180.00)



$$\frac{K_1 A}{L} (4T_0 - T) + \frac{K_2 A}{L} (2T_0 - T) = \frac{mSdT}{dt}$$

$$\frac{KA}{L} (4T_0 - T + 4T_0 - 2T) = \frac{mSdT}{dt}$$

$$\int_{T_0}^T \frac{dT}{8T_0 - 3T} = \frac{KA dt}{mSL}$$

$$-\frac{1}{3} \ln \left(\frac{8T_0 - 3T}{5T_0} \right) = \frac{KA t}{mSL}$$

$$-\frac{1}{3} \ln \frac{8T_0 - 3T}{5T_0} = \ln 2$$

$$\frac{8T_0 - 3T}{5T_0} = \frac{1}{8}$$

$$64T_0 - 24T = 5T_0$$

$$T = \frac{59}{24} T_0 = 59 \times 20$$

$$= 1180 \text{ K}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

10. Ans (A)

$$(P) K = \frac{K_A}{2} = \frac{0.693}{80} \text{ s}^{-1}$$

$$r = [A]_0 = \frac{0.693}{80} \times 0.8$$

$$= 6.93 \times 10^{-3} \text{ M/s}$$

$$(Q) r = \frac{-(4-2)}{40 \times 5} \times 10^{-2} \text{ M/s}$$

$$r = \frac{r_A}{2} = 5 \times 10^{-3} \text{ M/s}$$

$$(R) r = K[A]$$

$$= \frac{0.693}{80} \times 0.04$$

$$= 3.465 \times 10^{-4} \text{ M/s}$$

$$(S) KA = \frac{1}{t_{1/2}} \times \frac{1}{[A]_0} = \frac{1}{40} \times \frac{1}{4/5}$$

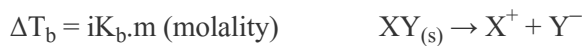
$$r = \frac{5}{40 \times 4 \times 2} \times (0.08)^2 = 2.5 \times 10^{-5} \text{ M/s}$$

PART-2 : CHEMISTRY

SECTION-II

4. Ans (1.00)

Elevation in boiling point Assuming ionic compound to be xy



$$0.52 = i \cdot 0.52 \times \frac{\text{moles}}{1 \text{ kg water (solvent)}}$$

i = Van'toff factor

$$\text{moles of compound} = \frac{1}{2} \text{ mole} = 2$$

$$\frac{1}{2} \text{ mole compound} = 3 \times 10^{23} \text{ molecules as}$$

100% dissociation

1 unit cell like fcc (NaCl) contains 4 moles of ionic compound

$$\text{So, number of unit cells} = \frac{3 \times 10^{23}}{4}$$

$$r^- = \text{radius of anion} = 2 \text{ \AA}$$

$$r^+ = \text{radius of cation} = r^+ \text{ \AA}$$

$$\text{edge length} = 2(r^- + r^+) = 2(2 + r^+) \text{ \AA}$$

$$\text{Volume of solid cube} = (2.532)^3 \text{ cm}^3$$

Volume of all the unit cell =

$$\frac{3}{4} \times 10^{23} \times [2(2 + r^+)]^3$$

No. of unit cell

$$21.6 = 10^{23} \times (2(2 + r^+))^3$$

$$6 \times 10^{-8} \text{ cm} = 2(2 + r^+) \text{ \AA}$$

$$6 \times 10^{-8} \text{ cm} = 2(2 + r^+) 10^{-8} \text{ cm}$$

7. Ans (2.00)

(b), (c)

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,D)

$$I = \int \frac{dx}{a + b \left(\frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \right)}$$

$$\text{Put } \tan \frac{x}{2} = t \Rightarrow \frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$I = \int \frac{2dt}{(a+b) + (a-b)t^2} = 2 \int \frac{dt}{\lambda_1 + \lambda_2 t^2}$$

obviously $\lambda_1 > 0$

C-1 If $\lambda_2 > 0$

$$\frac{2}{\sqrt{\lambda_1 \lambda_2}} \tan^{-1} \sqrt{\frac{\lambda_2}{\lambda_1}} \cdot t + c$$

$$= \frac{2}{\sqrt{\lambda_1 \lambda_2}} \tan^{-1} \left(\sqrt{\frac{\lambda_2}{\lambda_1}} \tan \frac{x}{2} \right) + c$$

C-2 If $\lambda_2 = 0$

$$I = 2 \int \frac{dt}{\lambda_1} = \frac{2}{\lambda_1} \tan \frac{x}{2} + c$$

C-3 If $\lambda_2 < 0$

$$I = 2 \int \frac{dt}{\lambda_1 - (-\lambda_2)t^2}$$

$$I = \frac{1}{\sqrt{-\lambda_1 \lambda_2}} \log_e \left| \frac{\sqrt{\lambda_1} + \sqrt{-\lambda_2} \tan \frac{x}{2}}{-\sqrt{\lambda_1} + \sqrt{-\lambda_2} \tan \frac{x}{2}} \right| + c$$

2. Ans (A,B,C,D)

$$f''(x) = \lambda x$$

$$f'(x) = \frac{\lambda x^2}{2} + c \Rightarrow f'(1) = 0 \Rightarrow c = -\frac{\lambda}{2}$$

$$f'(x) = \frac{\lambda}{2}(x^2 - 1)$$

$$f(x) = \frac{\lambda}{2} \left(\frac{x^3}{3} - x \right) + c$$

$$f(0) = c = 1$$

$$f(x) = \frac{\lambda}{2} \left(\frac{x^3}{3} - x \right) + 1$$

$$f(1) = 2 \Rightarrow \lambda = -3$$

$$f(x) = \frac{-x^3}{2} + \frac{3x}{2} + 1$$

$$(A) f(x) - f(-x) = 6x - 10$$

$$\Rightarrow -x^3 + 3x = 6x - 10$$

$$x^3 + 3x - 10 = 0$$

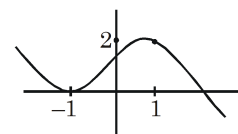
(Number of solution is 1 as the function is increasing)

$$(B) g(x) = \frac{x^3}{2} - \frac{3x}{2} - 1$$

$$g'(x) = \frac{3x^2}{2} - \frac{3}{2} \Rightarrow \frac{3}{2}(x-1)(x+1)$$

increasing of $g(x)$: $x \in (-\infty, -1] \cup [1, \infty)$

$$(C) f(x) = \frac{-x^3}{2} + \frac{3x}{2} + 1$$



$$f'(x) = -\frac{3}{2}(x-1)(x+1)$$

$$(D) p(\theta) = f(\sin\theta) \Rightarrow p(\theta) \in [0, 2]$$

3. Ans (A,B,C)

$$\sin x + \sin y = 1$$

$$\Rightarrow \cos x + \cos y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{\cos x}{\cos y}$$

$$\Rightarrow -\sin x - (\sin y) \left(\frac{dy}{dx} \right)^2 + (\cos y) \frac{d^2y}{dx^2} = 0$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{\sin x + \sin y \left(-\frac{\cos x}{\cos y} \right)^2}{\cos y}$$

$$= \frac{\cos^2 y \cdot \sin x + \sin y \cdot \cos^2 x}{(\cos y)^3}$$

$$\text{Put } \sin x = t, \sin y = 1 - t, \cos y = \sqrt{2t - t^2}$$

$$\lim_{x \rightarrow 0} \alpha \frac{d^2y}{dx^2} = \lim_{t \rightarrow 0} \frac{t^\alpha [(2t - t^2)t + (1 - t)(1 - t^2)]}{(\sqrt{2t - t^2})^3}$$

$$\text{exists} \Rightarrow \alpha \geq \frac{3}{2}$$

4. Ans (B,C)

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{\substack{x \rightarrow 1^+ \\ n \rightarrow \infty}} \frac{2 + \frac{x}{x^{2n}} + \frac{2}{x^{2n}}}{1 - \frac{x^2}{x^{2n}} + \frac{3}{x^{2n}}} = 2$$

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{\substack{x \rightarrow 1^- \\ n \rightarrow \infty}} \frac{2x^{2n} + x + 2}{x^{2n} - x^2 + 2}$$

$$= \lim_{x \rightarrow 1^-} \frac{x + 2}{-x^2 + 2} = \frac{3}{1} = 3$$

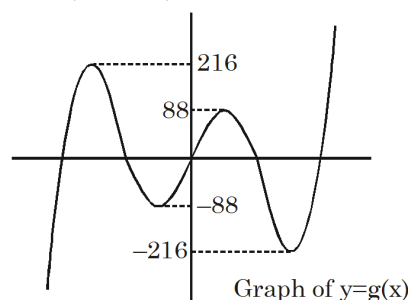
$\therefore \text{LHL} \neq \text{RHL}$

$\therefore$ Discontinuous at $x = 1$

Similarly

$$\lim_{x \rightarrow 2^+} f(x) = 2; \lim_{x \rightarrow 2^-} f(x) = 2$$

5. Ans (A,B,C)



$$\text{Let } g(x) = 3x^5 - 50x^3 + 135x$$

$$g'(x) = 0 \text{ for } x = \pm 1, \pm 3$$

$$g(1) = 88; g(-1) = -88; g(3) = -216; g(-3) = 216$$

Now consider $g(x) = a$

6. Ans (C,D)

$$I = \int \frac{(x^3 + 1)(x^3 - 2)}{x^3 \cdot x^2 \sqrt{\left(\frac{x^3 + 1}{x^2}\right)^2 + 3}} dx = \int \frac{\left(x + \frac{1}{x^2}\right) \left(1 - \frac{2}{x^3}\right)}{\sqrt{\left(x + \frac{1}{x^2}\right)^2 + 3}} dx$$

$$\text{Put } \left(x + \frac{1}{x^2}\right)^2 + 3 = t^2$$

$$2 \left(x + \frac{1}{x^2}\right) \left(1 - \frac{2}{x^3}\right) dx = 2t dt$$

$$\therefore I = \int \frac{t dt}{t} = t + C = \sqrt{\left(x + \frac{1}{x^2}\right)^2 + 3} + C$$

$$f(x) = \sqrt{\left(x + \frac{1}{x^2}\right)^2 + 3}$$

$$f(1) = \sqrt{7}$$

$$f(-1) = \sqrt{3} \text{ \& } f(2) < f(3)$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A)

$$(I) \phi(x) = 3f\left(\frac{x^2}{3}\right) + f(3 - x^2)$$

$$\phi'(x) = 2x \left(f'\left(\frac{x^2}{3}\right) - f'(3 - x^2) \right)$$

$$\text{Now } \frac{x^2}{3} - (3 - x^2) = \frac{4x^2}{3} - 3 > 0$$

$$\Rightarrow x < -\frac{3}{2}, x > \frac{3}{2}$$

$$\frac{4x^2}{3} - 3 < 0 \Rightarrow -\frac{3}{2} < x < \frac{3}{2}$$

$$f'\left(\frac{x^2}{3}\right) - f'(3 - x^2) \text{ is positive in } \left(-3, -\frac{3}{2}\right)$$

$$\text{and } \left(\frac{3}{2}, 4\right) \text{ and negative in } \left(-\frac{3}{2}, \frac{3}{2}\right) \text{ as } f'$$

(x) is increasing ($\because f''(x) > 0$)

$$\therefore \phi'(x) \text{ is negative in } \left(-3, -\frac{3}{2}\right), \left(0, \frac{3}{2}\right)$$

$$\text{i.e. decreasing in } \left[-3, -\frac{3}{2}\right], \left[0, \frac{3}{2}\right] \text{ and}$$

$$\text{increasing in } \left[\frac{3}{2}, 4\right], \left[-\frac{3}{2}, 0\right]$$

$$(II) g(x) = (a^2 - 3a + 2)\cos\frac{x}{2} + (a-1)x + \sin 1$$

$$g'(x) = -\frac{1}{2}(a^2 - 3a + 2)\sin\frac{x}{2} + (a-1)$$

$$= (a-1)\left(1 - \frac{1}{2}(a-2)\sin\frac{x}{2}\right)$$

$$g'(x) = 0 \text{ when } a = 1 \text{ or } \sin\frac{x}{2} = \frac{2}{a-2}, a \neq 2$$

$$\therefore \left|\frac{2}{a-2}\right| < 1 \Rightarrow a \in (-\infty, 0) \cup (4, \infty)$$

$\Rightarrow$ It possess critical points

if $a \in (-\infty, 0] \cup [4, \infty) \cup \{1\}$

8. Ans (C)

(P) Given equation holds good for $x = -1$ only

$$(Q) a = \lim_{x \rightarrow \infty} 5\left(2\left(\frac{3}{5}\right)^x + 3\left(\frac{4}{5}\right)^x + 4\right)^{1/x} = 5(0+0+4)^0 = 5$$

$$b = \lim_{x \rightarrow (-\infty)} 3\left(2 + 3\left(\frac{4}{3}\right)^x + 4\left(\frac{5}{3}\right)^x\right)^{\frac{1}{x}} = 3$$

(R) At $x = n, n \in \mathbb{I}$

$$\text{LHL} = n \sin(\pi(2n - (n-1)))$$

$$= n \sin((n+1)\pi) = 0$$

$$\text{RHL} = f(n) = n \sin n\pi = 0$$

$\therefore f(x)$ is always continuous

$$(S) \lim_{x \rightarrow 0} 3 \left(\frac{e^{x/3} - \frac{x}{3} - 1}{\frac{x^2}{9} \cdot 9} \right) = \frac{3}{9} \cdot \frac{1}{2} = \frac{1}{6}$$

9. Ans (A)

$$(P) g'(0) = f'(0) \cdot (-1 + 0 + f(0)) = 0$$

$$(Q) \text{ take log } \Rightarrow y(1 + \ln x) = x \\ \Rightarrow y = \frac{x}{1 + \ln x}$$

Now differentiate

$$(R) -1 \leq x \leq 2 \Rightarrow -2 \leq 2x \leq 4$$

$\therefore$ point of discontinuity

when $2x = -1, 0, 1, 2, 3, 4$

$$\text{i.e. } x = -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, 2$$

$$(S) I = e^{\lim_{x \rightarrow 0} (1 + \tan x - 1)} \cdot \frac{1}{\sin x} = e^1$$

10. Ans (B)

$$\sin^{-1}x + \cos^{-1}ay + \cos^{-1}(bxy) = \frac{\pi}{2}$$

$$\cos^{-1}(ay) + \cos^{-1}(bxy) = \cos^{-1}x$$

$$\cos(\cos^{-1}(ay) + \cos^{-1}(bxy)) = \cos(\cos^{-1}x)$$

$$(ay)(bxy) - \sqrt{1-a^2y^2} \cdot \sqrt{1-b^2x^2y^2} = x \quad \dots(1)$$

$$(P) a = 1, b = 2$$

$$\Rightarrow (y)(2xy) - \sqrt{1-y^2} \cdot \sqrt{1-4x^2y^2} = x$$

$$\Rightarrow (2xy^2 - x)^2 = (1-y^2)(1-4x^2y^2)$$

$$\Rightarrow (1-x^2)(1-y^2) = x^2y^2$$

$$(Q) a = 2, b = 1$$

$$\Rightarrow (2xy^2 - x)^2 = (1-4y^2)(1-x^2y^2)$$

$$\Rightarrow x^2 + 4y^2 = 1 + 3x^2y^2$$

$$(R) a = 0, b = 1 \Rightarrow -\sqrt{1-x^2y^2} = x$$

$$\Rightarrow x^2 = 1 - x^2y^2$$

$$\Rightarrow 2x = -x^2 2yy' - 2xy^2$$

$$\Rightarrow y' = -\frac{y^2 + 1}{xy}$$

$$(S) \text{ if } a = b \quad x^2 + a^2y^2 - x^2y^2a^2 - 1 = 0$$

$$\Rightarrow x = \pm 1 \text{ or } y = \pm \frac{1}{a}$$

$$\Rightarrow \text{area of rectangle} = 2 \cdot \frac{2}{a} = 1$$

$$\Rightarrow a = 4 = b$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.50)

$$\int \frac{(x^2 - 1) dx}{x^3 \sqrt{3x^4 + 2x^2 - 1}} = \int \frac{(x^2 - 1) dx}{x^5 \sqrt{3 + \frac{2}{x^2} - \frac{1}{x^4}}}$$

$$\int \frac{\left(\frac{1}{x^3} - \frac{1}{x^5}\right) dx}{\sqrt{3 + \frac{2}{x^2} - \frac{1}{x^4}}} \quad t = 3 + \frac{2}{x^2} - \frac{1}{x^4}$$

$$\Rightarrow \frac{dt}{dx} = \frac{-4}{x^3} + \frac{4}{x^5}$$

$$-\frac{1}{4} \int \frac{dt}{\sqrt{t}} = -\frac{1}{2} \sqrt{t} + C$$

$$f(x) = -\frac{1}{2} \sqrt{3 + \frac{2}{x^2} - \frac{1}{x^4}},$$

$$\lambda = -\frac{\sqrt{3}}{2} = \lim_{x \rightarrow \infty} f(x)$$

2. **Ans (3.20)**

$$f(x) = (x-1)^{2019} + (x-2)^{2019} + (x-3)^{2019} + \dots + (x-2019)^{2019}$$

$f(x)$ is an increasing function so it has only one real root

$$x = \frac{1+2019}{2} = \frac{2020}{2} = 1010$$

$$\frac{\lambda}{315} = \frac{1010}{315} = 3.20$$

3. **Ans (2.00)**

$$\tan(\cos^{-1}(\sin(\cos^{-1}x)))$$

$$= \tan\left(\frac{\pi}{2} - \sin^{-1}\left(\sin\left(\frac{\pi}{2} - \sin^{-1}x\right)\right)\right)$$

$$= \cot(\sin^{-1}(\cos(\sin^{-1}x)))$$

$$\Rightarrow f(x) = 1$$

$$\sum_{r=0}^{\infty} \frac{1}{2^r} = \frac{1}{1 - \frac{1}{2}} = 2$$

4. **Ans (0.50)**

$$\lim_{n \rightarrow \infty} \prod_{r=1}^{r=n-1} \left(1 - \frac{1}{(r+1)^2}\right)$$

$$\prod_{r=1}^{r=n-1} \frac{(r+1)^2 - 1}{(r+1)^2} = \prod_{r=1}^{r=n-1} \frac{r}{(r+1)} \cdot \frac{(r+2)}{r+1}$$

$$= \prod_{r=1}^{r=n-1} \frac{r}{r+1} \cdot \prod_{r=1}^{r=n-1} \frac{r+2}{r+1}$$

$$= \frac{1}{n} \cdot \frac{(n+1)}{2}$$

$$\lim_{n \rightarrow \infty} \prod_{r=1}^{r=n-1} \left(1 - \frac{1}{(r+1)^2}\right)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \cdot \frac{(n+1)}{2} = \frac{1}{2}$$

5. **Ans (9.00)**

$$f'(0^-) = \lim_{h \rightarrow 0} \frac{\left(\frac{(0-h)(\ln(1-h) + \ln(1+h))}{(\sec h - \cosh)} - 0\right)}{-h}$$

$$f'(0^-) = \lim_{h \rightarrow 0} \frac{\ln(1-h^2)}{(\sec h - \cosh)}$$

$$= \lim_{h \rightarrow 0} \frac{\ln(1-h^2)}{-h^2} \cdot \frac{\cosh}{\frac{1-\cosh}{h^2}} \times \frac{1}{(1+\cosh)} \times \frac{-h^2}{h^2}$$

$$= -1$$

$$f'(0^+) = (p^2 - 3p - 1)\cos 0 + 2 \times 0$$

$$= (p^2 - 3p - 1)$$

$$f'(0^-) = f'(0^+)$$

$$p^2 - 3p - 1 = -1$$

$$p = 0, 3 \Rightarrow p_1 = 0 \text{ \& } p_2 = 3$$

6. **Ans (0.20)**

Let

$$P(x) = a(x-1)^4 + b(x-1)^3 + c(x-1)^2 + d(x-1) + e$$

$$P(1) = e = 1$$

$$P'(1) = d = 0$$

$$P''(1) = 2c = 2 \Rightarrow c = 1$$

$$P'''(1) = 3.2.b = -12 \Rightarrow b = -2$$

$$P''''(1) = 4.3.2.1.a = 24 \Rightarrow a = 1$$

$$\therefore P(x) = (x-1)^4 - 2(x-1)^3 + (x-1)^2 + 1$$

$$\therefore P(2) = 1 \text{ \& } P(3) = 5$$

$$\therefore \frac{P(2)}{P(3)} = \frac{1}{5} = 0.2$$

7. **Ans (8.00)**

Eliminating t will give

$$y^2(x-1) = 1$$

equation of tangent at $(2, 1)$ is $x + 2y = 4$

solving with curve,

$$\text{we get } x = 5 \text{ and } y = -\frac{1}{2}$$

$$Q = \left(5, -\frac{1}{2}\right) \text{ or } |PQ| = \frac{3\sqrt{5}}{2}$$

8. **Ans (3.75)**

$$f\left(\frac{2x-3}{x-2}\right) = 5x-2$$

$$\frac{2x-3}{x-2} = f^{-1}(5x-2)$$

$$x = 3 \quad f^{-1}(13) = 3$$